

Dark matter as a geometric scalar: the Aether-Scalar-Tensor host for a cosmological-constant acceleration scale, and its price

C. Zimmerman¹

¹*Independent*

(Dated: June 2026)

The hypothesis that galactic “dark matter” is a modification of dynamics with an acceleration scale fixed by the cosmological constant, $a_0 = c^2 \sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m s}^{-2}$, requires a covariant host that survives the gravitational-wave-speed bound, lenses correctly, and is consistent with the cosmic microwave background. We argue that the Aether-Scalar-Tensor theory (AeST) is that host, and account honestly for what it buys and what it costs. AeST is built from a single metric, a unit-timelike vector field (the “aether”), and a shift-symmetric scalar; it is ghost-free, propagates tensor modes at the speed of light in all backgrounds—surviving GW170817 where the earlier bimetric TeVeS did not—and it yields two structural results we rely on: gravitational lensing has no slip ($\Phi = \Psi$, so lensing mass equals dynamical mass) and the acceleration scale a_0 is identically absent from linear cosmological perturbation theory. Crucially, in AeST the galactic scale a_0 and the cosmological dark-matter-like component live in *orthogonal* sectors of one free function: a_0 enters a forced spatial-gradient term, while the cosmic “dust” that drives the third acoustic peak is a shift-symmetric k-essence mode whose amplitude is a free integration constant. AeST therefore fits the full Planck spectrum—but at the price of one early-universe density parameter that $a_0 = \Lambda$ does not supply: the unification of the dark sectors is galactic, not cosmological. We name the full price (a second density Ω , a free cluster mass scale, an underived $\mathcal{O}(1)$ coefficient Z) rather than hide it, and conclude that the geometric-scalar reading of dark matter is viable and falsifiable, at one honest extra number.

I. THE THESIS AND THE HOST

If galaxy rotation curves reflect modified dynamics rather than unseen particles, and if the modifying acceleration scale is set by the cosmological constant via $a_0 = c^2 \sqrt{\Lambda/32\pi}$ [1, 2], then the proposal lives or dies on a relativistic completion. Three things have historically killed relativistic MOND: the speed of gravitational waves (the bimetric Tensor-Vector-Scalar theory, TeVeS [3], predicted $c_T \neq c$ and was excluded by GW170817 [4]); the under-prediction of gravitational lensing; and the cosmic microwave background (CMB), whose third acoustic peak demands a clustering, pressureless component that a force law does not provide. The Aether-Scalar-Tensor theory (AeST) of Skordis and Złośnik [2] clears all three. This paper takes AeST as the host for the cosmological-constant acceleration scale and accounts, both ways, for what that costs.

II. THE GEOMETRIC STRUCTURE

AeST is economical. Its field content is one metric $g_{\mu\nu}$, one unit-timelike vector A_μ with $A^\mu A_\mu = -1$ (the “aether,” satisfying $\nabla_\mu A^\mu = 3H$ on a Friedmann background), and one shift-symmetric scalar ϕ [2]. Matter couples only to $g_{\mu\nu}$, so the weak equivalence principle holds exactly; the vector A_μ selects a preferred cosmic rest frame, which is precisely the structure a de Sitter–Unruh reading of inertia requires. The vector is the descendant of Einstein-Aether theory [5]; the disformal role it plays in fixing lensing is inherited from TeVeS but re-

alised here on a *single* metric, with a kinetic term engineered so that the tensor sector propagates at $c_T = c$ in every background. This is why AeST survived the neutron-star merger that excluded TeVeS.

III. TWO STRUCTURAL RESULTS

Two consequences of the A_μ structure are not tuned but forced, and we use both.

a. No-slip lensing. Because the time component of the aether tracks the metric, $A^0 \sim \sqrt{-g^{00}}$, the two weak-field potentials are equal, $\Phi = \Psi$ [6]. Light deflection ($\propto \Phi + \Psi$) and dynamics ($\propto \Phi$) therefore read the same enhanced potential, and the lensing mass equals the dynamical mass for any source. This repairs the historic under-lensing of non-relativistic MOND, and it means that cluster strong+weak lensing, the Bullet cluster offset, and Einstein radii are not independent dark-matter detections but restate the cluster dynamical residual.

b. The acceleration scale is CMB-invisible. On a Friedmann background the spatial-gradient invariant of the scalar vanishes identically, so the term that carries a_0 contributes nothing to the linear perturbation equations [2]. The galactic acceleration scale is therefore *absent* from the CMB transfer functions: it can neither help nor hurt the acoustic peaks. Whatever fits the CMB in AeST does so independently of a_0 .

IV. THE DECOUPLING: TWO ORTHOGONAL SECTORS

The central structural fact, and the source of both the success and the cost, is that a_0 and the cosmological dark-matter-like component occupy *different* pieces of the AeST free function $\mathcal{F}(Y, Q)$. The galactic scale enters only the spatial-gradient sector, through a forced fractional-power term $\mathcal{F} \supset [2\lambda_s/3(1+\lambda_s)a_0]Y^{3/2}$ as $|\nabla\phi| \rightarrow 0$ —the $3/2$ power is the square-root force law, and it is here, and only here, that a_0 appears [2]. The cosmological component lives in the orthogonal $\mathcal{K}(Q)$ sector: a shift-symmetric k-essence scalar with a minimum at $Q_0 \neq 0$ behaves, once integrated, as pressureless dust, $\bar{\rho} \propto a^{-3}$, plus a cosmological constant. The dust amplitude is set by an integration constant I_0 , $8\pi\tilde{G}\bar{\rho}_0 = Q_0 I_0$; in the authors’ words the density “is not (classically) predicted.” Galaxies and the early universe are thus controlled by independent slots of one function.

V. THE CMB AT ONE EXTRA NUMBER

AeST reproduces the full Planck temperature and polarization spectra and the linear matter power spectrum to the quality of Λ CDM—the first relativistic MOND theory to do so [2]. The mechanism is the $\mathcal{K}(Q)$ dust mode: a pressureless density $\propto a^{-3}$ that drives the acoustic peaks exactly as cold dark matter does. We confirmed independently, with a standard Boltzmann code, that the third-peak height is a direct measure of this clustering density: the ratio of third- to second-peak amplitude rises monotonically with the physical cold density, from $\simeq 0.53$ for a baryon-only universe to $\simeq 0.98$ for the observed $\Omega_c h^2 \simeq 0.12$, and a pure modified-inertia universe (in which a_0 adds nothing to the linear physics) cannot reach the observed value by any tuning.

The price is explicit. The dust amplitude $I_0 \sim \Omega_c h^2$ is a free integration constant, not delivered by $a_0 = c^2\sqrt{\Lambda/32\pi}$. The acceleration scale fixes the dark-energy face ($\Omega_\Lambda = 0.685$ for the measured Λ), but the dark-energy and the CMB dust densities stand in ratio $\rho_{\text{DE}}/\rho_c \simeq 2.6$ today and redshift differently (constant versus $(1+z)^3$, diverging by recombination), so one number cannot set both. The slogan “two dark sectors, one number” holds for galaxy dynamics ($a_0 \leftrightarrow \Lambda$ generates the radial-acceleration and Tully–Fisher relations) but breaks at the CMB, which requires a second, independently tuned density. The CMB is not evidence against the framework—AeST fits it—nor is it a flaw in the data; it is a cost in unification economy of exactly one early-universe parameter.

VI. THE CLUSTER RESIDUAL

The same logic clarifies clusters. The dynamical residual at R_{500} , $\eta \simeq 2.3$ on the eROSITA eRASS1 sample [7],

is shared with all modified dynamics and is not cured by $a_0 = \Lambda$; by the no-slip identity the cluster lensing residual is the *same* number, not an independent one. Three honest points temper it. First, the eRASS1 masses are weak-lensing-calibrated, and the lensing masses run $\sim 110\%$ above the hydrostatic and kinematic masses (which agree with each other), bracketing the true residual to $\eta \in [\sim 1, 2.3]$; direct microcalorimeter measurements of non-thermal pressure and the projection bias of lensing masses pull the convergent estimate to $\eta \sim 1.3$ – 1.9 . Second, the framework’s own lower a_0 makes the residual $\sim 13\%$ worse ($\eta \propto a_0^{-1/2}$), so it is not a convention artifact. Third, the same $\mathcal{K}(Q)$ dust that fits the CMB is the leading candidate to supply the residual at clusters, and the AeST scalar mass term yields a right-signed enhancement above MOND at cluster scales [8]—but the mass term is non-monotone, is squeezed between galaxy- and cluster-lensing constraints on the same parameter [9], and any such cure introduces its own density scale, not $a_0 = \Lambda$. The cluster residual is real, soft, systematic-limited, and uncured as a unification win.

VII. THE PRICE, NAMED

We state the full cost of the geometric-scalar reading, rather than bury it. Beyond $a_0 = c^2\sqrt{\Lambda/32\pi}$ the AeST host requires: (i) one early-universe density, the integration constant $I_0 \sim \Omega_c h^2$, that drives the CMB and is the leading cluster candidate—orthogonal to a_0 ; (ii) a free cluster-scale mass μ in the scalar sector, constrained from opposite directions by galaxy and cluster lensing; (iii) the shape of the free function $\mathcal{F}(Y, Q)$, chosen to fit; and (iv) the dimensionless coefficient $Z = \sqrt{32\pi/3} \simeq 5.79$ in $a_0 = cH_\Lambda/Z$, which the de Sitter geometry fixes only to $\mathcal{O}(1)$ and which we do not assert as derived. What the geometry *does* fix, with no freedom, is the form and scale: $a_0 \sim c^2\sqrt{\Lambda}$ is the Gibbons–Hawking horizon acceleration, and the appearance of a_0 in a $Y^{3/2}$ term reproduces the square-root force law. Form and scale: forced. Coefficient and cosmological unification: posited.

VIII. CONCLUSION

The proposition that dark matter is an illusion—a geometric scalar field in a Lorentz-violating completion of general relativity, with an acceleration scale set by the cosmological constant—has, in AeST, a host that is ghost-free, gravitational-wave-safe, lenses without slip, and fits the cosmic microwave background. That is a substantive position, not a slogan. It is also not free: the cosmological dark-matter density is one parameter distinct from the galactic $a_0 = \Lambda$ unification, the cluster residual is real if soft, and the coefficient Z is an honest posit. The geometric-scalar reading should be judged on those terms—one extra early-universe number, and a falsifiable, distinctive prediction the particle picture cannot

make: a redshift-declining $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$.

ACKNOWLEDGMENTS

The Boltzmann third-peak recomputation, the eRASS1 residual at the framework’s own a_0 , and the

AeST sector accounting were performed and cross-checked numerically; the AeST identities are quoted from the primary literature. Both-ways audit ledgers are archived with this work.

-
- [1] M. Milgrom, *Astrophys. J.* **270**, 365 (1983).
 - [2] C. Skordis, T. Złóśnik, *Phys. Rev. Lett.* **127**, 161302 (2021) [AeST].
 - [3] J. D. Bekenstein, *Phys. Rev. D* **70**, 083509 (2004) [TeVSe].
 - [4] B. P. Abbott *et al.* (LIGO/Virgo, Fermi, INTEGRAL), *Astrophys. J.* **848**, L13 (2017).
 - [5] T. Jacobson, D. Mattingly, *Phys. Rev. D* **64**, 024028 (2001) [Einstein-Aether].
 - [6] P. Verwayen, C. Skordis, C. Złóśnik, *Mon. Not. R. Astron. Soc.* **531**, 272 (2024).
 - [7] E. Bulbul *et al.*, *Astron. Astrophys.* **685**, A106 (2024) [eRASS1].
 - [8] A. Duraković, C. Skordis (2024) [AeST scalar mass term in clusters].
 - [9] T. Mistele, S. McGaugh, S. Hossenfelder, *Astron. Astrophys.* **676**, A100 (2023).